

PCCST501 - COMPUTER NETWORKS

Assignment 3

Module 3 - Data-Link Layer, Wireless LANs & Mobile IP

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Course Outcome	CO3

LAN, Ethernet & Wireless Network Investigation

This report answers all ten questions in the supplied assignment brief. The explanations and terminology are aligned with the supplied Data-Link Layer and Wireless Networks & Mobile IP course PDFs. A real Wireshark capture is still required for Question 3; no MAC addresses are fabricated.

1. Responsibilities of the Data-Link Layer; LLC vs MAC

The data-link layer provides **node-to-node communication**. A packet from the network layer may cross several different networks; at each link, the data-link layer is responsible for delivering the frame to the next node over that specific link.

The supplied text divides the layer into two sublayers. **Data Link Control (DLC)** deals with functions common to point-to-point and broadcast links, while **Media Access Control (MAC)** deals with issues specific to broadcast/shared links. DLC includes framing, flow control, and error control; MAC determines how stations obtain access to a shared medium.

Framing packages bits into recognizable frames. **Flow control** prevents a fast sender from overwhelming a receiver, and **error control** detects damaged/lost frames and coordinates retransmission. At the MAC level, an access method is needed when several stations share the same medium.

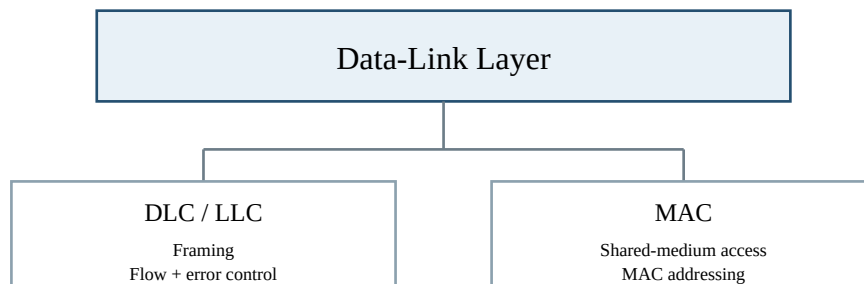


Figure 1. Functional division of the data-link layer (schematic).

2. Ethernet Frame Format and Purpose of Key Fields

An Ethernet frame is the data-link-layer unit used to carry data across an Ethernet link. The frame contains addressing/control fields and the payload supplied by the upper layer.

Destination MAC: identifies the link-layer destination to which the frame is addressed. **Source MAC:** identifies the transmitting interface. **Type/Length:** identifies the interpretation of the following field - commonly the upper-layer protocol in Ethernet II framing, or the payload length in the IEEE 802.3 interpretation. **FCS:** carries the frame-check value used by the receiver to detect transmission errors.

The data-link text explains the general purpose of framing: it separates one message from another and associates the frame with sender and destination addresses. Smaller frames can also make error and flow handling more efficient because an error need not force retransmission of an extremely large message.

Preamble	SFD	Destination MAC	Source MAC	Type / Length	Data / Pad	FCS
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Start of frame

End of frame

Figure 2. Ethernet frame layout - conceptual diagram for the assignment.

3. Wireshark Capture - Source and Destination MAC Addresses

The assignment requires at least five captured Ethernet frames. In Wireshark, select Ethernet frames in the packet list and read the **Source** and **Destination** addresses in the packet details. Record the values exactly as displayed in the capture.

The table is intentionally left for the student's real capture evidence. This is important because MAC addresses are device-specific and should not be invented for an academic lab report. Insert one readable Wireshark screenshot after the table showing the selected frame and its Ethernet header.

4. Switch Operation Using MAC Address Learning

A switch learns by inspecting the **source MAC address** of each incoming frame. It associates that address with the port on which the frame arrived, building a forwarding (MAC) table.

When a frame arrives, the switch examines the destination MAC address. If the destination is in the table, the switch forwards the frame only through the port associated with that address. If the destination is unknown, the frame is flooded out the other appropriate ports. When later frames reveal more source addresses, the table becomes increasingly complete and forwarding becomes selective.

Example: A sends a frame through Port 1, so the switch records A -> Port 1. B then sends through Port 2, so it records B -> Port 2. If A later sends to B, the known B -> Port 2 entry allows the switch to forward only to Port 2.

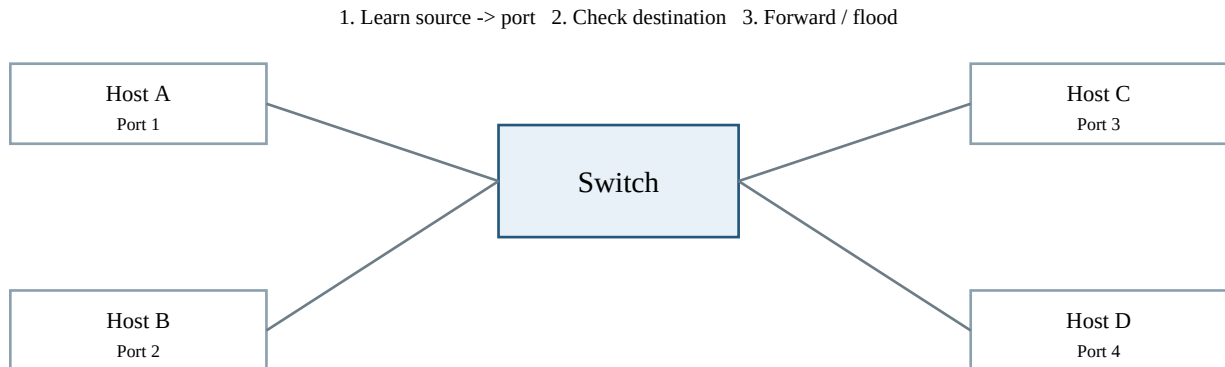


Figure 3. MAC address learning and selective forwarding (schematic).

5. Hub vs Switch vs Bridge

A **hub** operates as a simple shared-medium device: an incoming signal is repeated toward other ports, so all connected stations share the collision domain. A **bridge** connects LAN segments and makes forwarding decisions using link-layer addresses. A **switch** is a multiport LAN forwarding device that learns MAC addresses and can make per-port forwarding decisions.

The supplied Chapter 5 text identifies hubs, link-layer switches, and routers as connecting devices in the lower layers of the TCP/IP suite. In practical LANs, the switch is preferred because it provides selective forwarding and supports point-to-point full-duplex links.

Device	Forwarding behaviour	Collision domains	Typical use
Hub	Repeats received signals to other ports	Shared collision domain	Legacy/shared Ethernet
Bridge	Filters/forwards using link-layer addresses	Segments separated by bridge	Connecting LAN segments
Switch	Learns MAC addresses and forwards per port	Each port switched full-duplex link is separate collision domain	Modern Ethernet LANs

6. CSMA/CD and Why Its Importance Is Reduced in Switched Full-Duplex Ethernet

CSMA/CD means Carrier Sense Multiple Access with Collision Detection. A station first senses the medium. If the medium is idle, it transmits. If two stations transmit at the same time on a shared Ethernet medium, a collision can occur; the stations detect the collision and use a back-off process before retrying.

Its importance is reduced in modern switched Ethernet because the usual host-to-switch link is a separate **point-to-point full-duplex** connection. A host does not need to compete with other hosts for the same shared wire segment, so collisions are not the normal operating condition. CSMA/CD therefore remains mainly relevant to legacy shared Ethernet rather than ordinary switched full-duplex LANs.

7. CSMA/CA in IEEE 802.11 and Comparison with CSMA/CD

Wireless LANs use a shared air medium. The supplied wireless chapter explains that CSMA/CD is unsuitable because a wireless station generally cannot transmit and detect a collision at the same time, hidden stations may not hear one another, and signal fading can prevent a station from hearing a collision at a distant location.

CSMA/CA therefore focuses on avoiding collisions. In IEEE 802.11 DCF, a station senses the channel, waits for an interframe space (IFS), chooses a random contention-window back-off, and senses the channel during the waiting period. After successful reception, an acknowledgment is used. RTS/CTS can further help with hidden-station situations.

Difference: CSMA/CD detects collisions after transmission starts and then retries; CSMA/CA tries to reduce the probability of collision before transmission and uses acknowledgments to confirm reception.

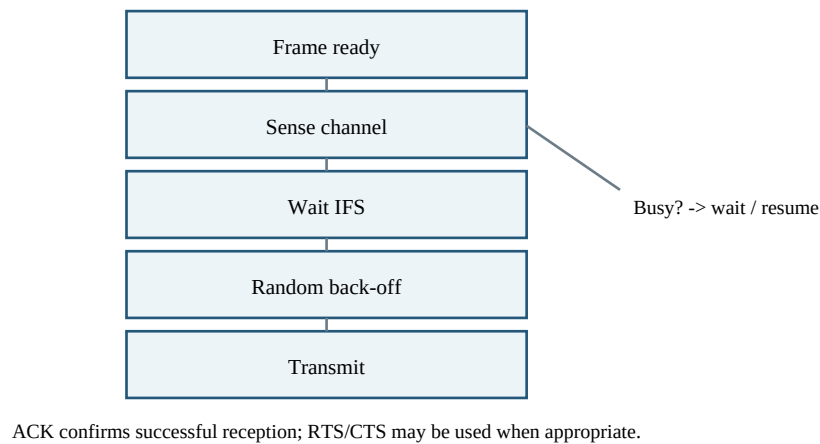


Figure 4. Simplified CSMA/CA sequence.

8. Small Wireless LAN Design for a College Laboratory

A practical laboratory WLAN should use an **infrastructure network** with an access point (AP). The supplied wireless chapter explains that an infrastructure WLAN connects wireless stations to a wired infrastructure through an AP. The AP therefore acts as the point where the wireless and wired environments meet.

Placement: position the AP near the centre of the laboratory, above major obstructions where practical, and away from large sources of interference. For a larger or crowded lab, use multiple APs with overlapping coverage designed deliberately rather than simply increasing transmit power.

SSID: use a meaningful identifier such as VJCET-Lab. **Channel:** choose channels to reduce co-channel interference; with multiple APs, use a planned channel allocation. **Addressing:** use a private subnet such as 192.168.10.0/24 with DHCP for student devices and reserved/static addresses for infrastructure devices. **Security:** use WPA2/WPA3, strong authentication credentials, a changed administrator password, and separate staff/student access where needed.

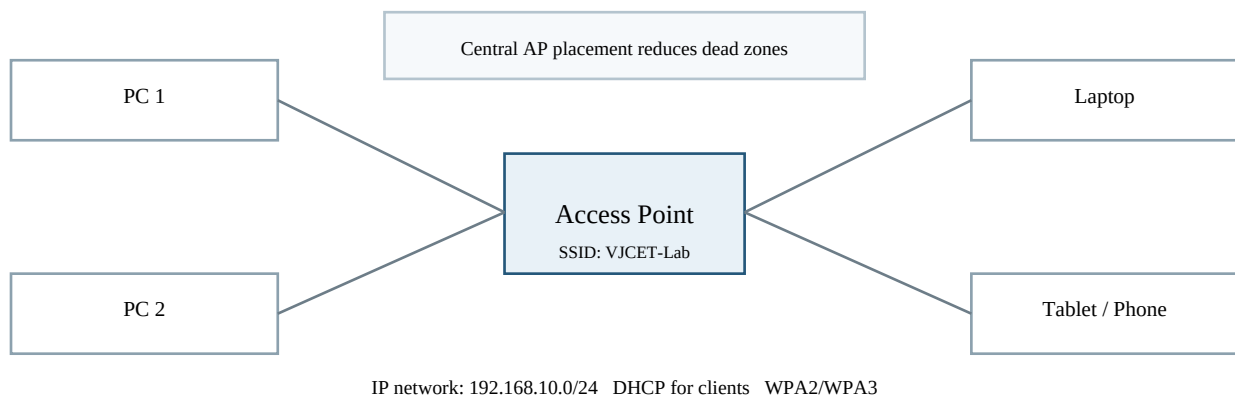


Figure 5. Proposed college laboratory WLAN (schematic).

9. Mobile IP and the Need for Mobility Support

A mobile host may move from one network to another. The central problem is that the host's network-layer location can change even though the host is still the same communicating device. If ordinary IP addressing is tied only to the original point of attachment, packets may no longer reach the host at its new location.

The supplied wireless chapter treats Mobile IP as IP support for mobile hosts and discusses **addressing, the phases of mobile access, registration, agents, and data transfer**. The mobile host identifies its current point of attachment, registers the relevant location information, and uses mobility-support mechanisms so that communication can continue while the host changes networks.

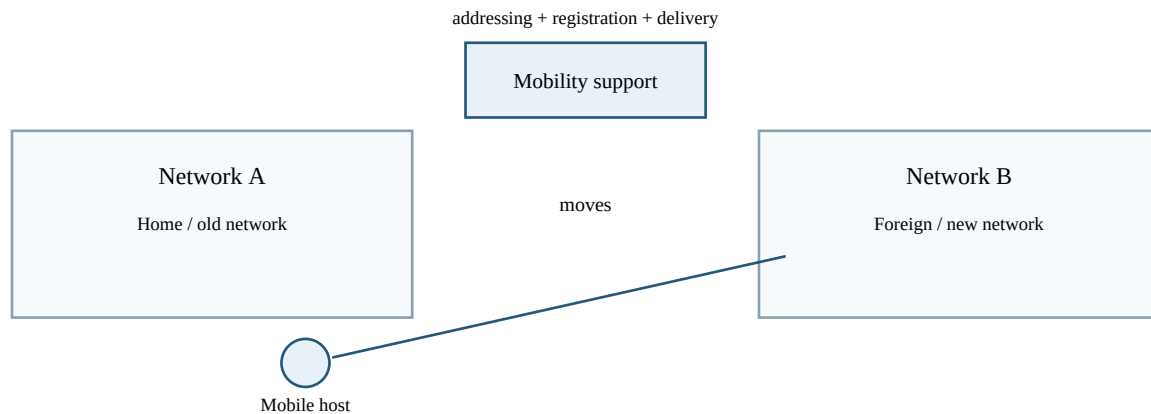


Figure 6. Conceptual Mobile IP movement between networks.

10. Packet-Level Comparison: Wired Ethernet vs IEEE 802.11 Wireless LAN

Both wired Ethernet and IEEE 802.11 provide link-layer delivery, but their frame headers reflect different media and access requirements. Ethernet frames use destination and source MAC addresses and include a Type/Length field and FCS. IEEE 802.11 introduces a more detailed MAC structure because wireless operation requires additional control and addressing information.

The supplied wireless chapter describes the IEEE 802.11 MAC architecture and shows DCF/CSMA/CA, interframe spaces, contention windows, acknowledgments, and RTS/CTS. In a packet-level investigation, the student should compare the fields actually visible in a capture and explain how the wireless frame supports the shared-air environment.

Feature	Wired Ethernet	IEEE 802.11 WLAN
Medium	Wired cable / switched link	Air / shared wireless medium
Destination/source MAC	Present	Present; additional address fields may be present depending on frame context
Type / control structure	Type/Length + Ethernet control fields	802.11 MAC control fields, including Frame Control and other fields
Access	Modern switched links are usually point-to-point; legacy shared links used CSMA/CD	DCF/CSMA/CA
Collision handling	Legacy collision detection and retransmission	Collision avoidance, back-off, acknowledgments; RTS/CTS may help
Capture evidence	Ethernet II / 802.3 details	802.11 MAC details

Conclusion

The data-link layer provides reliable node-to-node delivery through framing, addressing, flow/error control, and media-access mechanisms. Ethernet evolved from shared-medium operation toward switched full-duplex links, reducing the role of CSMA/CD. Wireless LANs retain a shared medium and therefore require CSMA/CA and additional control mechanisms. Mobile IP extends the idea of connectivity to hosts that change their network attachment point.

References

- Behrouz A. Forouzan, Data Communications and Networking - supplied Data-Link Layer: Wired Networks and Wireless Networks and Mobile IP course PDFs.
- PCCST501 - Computer Networks, Assignment 3, Module 3 - supplied assignment brief.

Submission evidence reminder: add five actual Wireshark frame records and readable screenshots before submission. The assignment brief explicitly requires Wireshark screenshots and a PDF report.